

Block 11c: build-variant and scanner review repairs

Date: 2026-09-28. Auditor: Codex /root. Isolated checkout: /private/tmp/kk715-11a. Immediate target: fork audit/715-11b-solana-runtime-20260928 at 51f99694f0cdde2151726a7fa252956a60ac1169 (PR #881). Code generation predecessor: b37a400ddfe7543910da8e50e49006e41455b3ed. Final containing head and hosted CI belong in the PR body. Main-worktree master template SHA256 at intake: 985cd8f5708e20ac34bf7442454b7ddb2271f5e9a2de048006dd75fb6f4c5556.

1. Scope and source reconciliation

This bounded unit extracts five unchanged paths from cumulative Block 11 after the first Copilot review 5333270562. It removes the broad ^deps/ scanner allowlist, places Pallas and ZIP-316 sources behind KK_ZCASH_PRIVACY, fixes emulator shared-library fallback paths to existing board sources, and removes the duplicate standalone zcash-crypto-unit target and its local RNG stub. Full firmware-unit retains the shared Zcash cases. The two exact public-gitlink no-git scanner ignores and report-authority test are inherited from approved prerequisite 11p, outside this adjacent diff. Product protocol, runtime handler and dependency pins do not change here.

Direct pins remain: code-signing-keys a6470bd8598e5e9a7bfc38bf139a5e5a616f05ec; deps/crypto/trezor-firmware 8a392f70a5d5575ece3dfb35f115d4a4b27f497c; deps/device-protocol 5fec9e6906a340be5eb3d795ec746769065b2db8; deps/googletest 7888184f28509dba839e3683409443e0b5bb8948; deps/python-keepkey aae89d378d889b3872696469fe363cf570e5ac4c; deps/qrenc/QR-Code-generator 6dfbfad5d9303ed190d1c3cb7bec34b565b6ce8; deps/sca-hardening/SecAESSTM32 71d356a1141624994cf613bd2d2583892e8e6d5a. Direct and nested submodules were initialized and checked clean before local builds.

2. Finding classes and verification

The scanner-sensitive value is a potential credential in a tracked dependency path; the observer is Gitleaks in CI. The old ^deps/ allowlist could suppress such a finding. In an isolated disposable deps/ fixture with a synthetic token, old configuration finds zero and fixed configuration finds one, with the value redacted. This is a configuration negative control, not a claim about an actual secret. Local Gitleaks is 8.30.0. A scan of a recursively populated local dependency tree also reports third-party fixture material outside the hosted clean superproject checkout; it is not used as the CI acceptance run. Hosted CI pins 8.30.1 and must pass on this exact PR head.

The build-sensitive state is product source composition. The old crypto source list included Pallas files before its Zcash option check, whereas the fixed full source graph contains Pallas and the Bitcoin-only graph excludes it. The old shared-library fallback referenced nonexistent emulator-relative strlcpy.c/strlcat.c; the fixed board-relative files exist and both compile with the pinned Linux Clang. A forced optional-dylib configure under the older Linux CMake stops at an unrelated cross-directory target-link ownership error before this target builds. The cumulative Windows cross-build is carried source evidence; this unit's hosted dylib build is the exact-head gate for the supported host. No Windows runtime execution is claimed.

The Zcash assertion is coverage preservation: the removed duplicate target does not remove the registered firmware suite. The full source graph builds unittests/firmware/zcash.cpp; 72/72 focused Zcash tests pass. Bitcoin-only excludes that product feature and has zero Zcash cases. The review's suggestion that all Zcash tests were absent is refuted by those executed cases; the redundant target itself was a valid cleanup. The source graph also shows no zcash-crypto-unit target after removal.

Local check	Full	Bitcoin-only	Outcome
Firmware native	669	143	All pass, zero failures/errors/skips
Crypto native	18	18	All pass, zero failures/errors/skips
Focused Zcash	72	0 by product	All full cases pass
Pallas source graph	Included	Excluded	Correct variant boundary
Scanner synthetic fixture	Old 0 / fixed 1	Same policy	Broad exemption removed

The deterministic sixteen-member 715-11c-controls-20260928.tgz archive has SHA256 41b4507721a651566a5705272cfabdc7a8f9f5a419e6602f1dc79933399a5e3. Its manifest binds fifteen raw XML/log/source-graph/control files and their hashes. A changed XML member is rejected. The related first-review control archive and every historical inline disposition remain available on cumulative PR #877.

3. Local and external checkpoint

The build-variant, Zcash coverage and scanner-configuration properties have current local assertions; native full and Bitcoin-only suites pass. The optional dylib fallback is verified by source-path and object compilation locally, with full target qualification delegated to exact-head hosted CI. Final post-edit preflight, hosted result, PDF/inventory check and Copilot delivery are recorded after report freeze in the PR body. This code-bearing unit has a fresh three-round review budget under the owner's consecutive split authority. No review has been requested here yet. Physical-device, merge and release acceptance remain separate.

4. Complete adjacent Git inventory

Inventory base: 51f99694f0cdde2151726a7fa252956a60ac1169. Code predecessor:

b37a400ddfe7543910da8e50e49006e41455b3ed; first containing report predecessor:

6b3e7643aa5ddef41b737adb161ec7017d9d534c. The table was regenerated after that first report commit grew to include its own final rows. Reproduce from the immutable base to the final head recorded in the PR body with `git diff --numstat 51f99694f0cdde2151726a7fa252956a60ac1169..FINAL_PR_HEAD`. Binary counts use Git's dash.

Added	Deleted	Path
2	3	.gitleaks.toml
3	8	deps/crypto/CMakeLists.txt
46	0	docs/release/audit-units/715-11c-build-review-repairs-20260928.md
-	-	docs/release/audit-units/715-11c-build-review-repairs-20260928.pdf
-	-	docs/release/audit-units/715-11c-controls-20260928.tgz
4	4	lib/emulator/CMakeLists.txt
2	13	unittests/crypto/CMakeLists.txt
0	21	unittests/crypto/emulator_random.c